

Daniel Vedensky

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Education

- 2025 **PhD. Statistics**, *University of Missouri*, Columbia, MO
Dissertation: *Bayesian Hierarchical Unit-Level Models for Longitudinal Survey Data Under Informative Sampling*
Advisor: Prof. Scott H. Holan
- 2022 **M.A. Statistics**, *University of Missouri*, Columbia, MO
- 2013 **B.A. Linguistics**, *Reed College*, Portland, OR

Experience

- 2025–Current **Postdoctoral Fellow**, *Simon Fraser University*, Burnaby, BC
Advisor: Prof. Donald Estep
- 2021–2025 **Graduate Research Assistant**, *University of Missouri*, Columbia, MO
Funded by U.S. Census Bureau Dissertation Fellowship, 8/23–2/25
- 2020–2021 **Graduate Teaching Assistant**, *University of Missouri*, Columbia, MO
- 2013–2016 **NLP Speech Scientist**, *Nuance Communications*, Seattle, WA

Publications

- Kawano[†], S., Vedensky[†], D., Dong, Q., Pawl, E., Wang, Q., Parker, P. A., Li, Z. R., and Holan, S. H. (2026). “Nonprobability Samples for Small Area Estimation: A Review and Comparative Simulation Study.” [arXiv:2608.13673](https://arxiv.org/abs/2608.13673).
- Vedensky, D., Parker, P. A., and Holan, S. H. (2025b). “Bayesian unit-level modeling of categorical survey data with a longitudinal design.” (Under revision). [arXiv:2502.15112](https://arxiv.org/abs/2502.15112).
- Vedensky, D., Parker, P. A., and Holan, S. H. (2025a). “Bayesian unit-level models for longitudinal survey data under informative sampling: an analysis of expected job loss using the Household Pulse Survey.” *Journal of the Royal Statistical Society Series A: Statistics in Society*. [doi:10.1093/jrssa/qnaf043](https://doi.org/10.1093/jrssa/qnaf043).
- Vedensky, D., Parker, P. A., and Holan, S. H. (2023). “A Look into the Problem of Preferential Sampling through the Lens of Survey Statistics.” *The American Statistician*, 77, 3, 313–322. [doi:10.1080/00031305.2022.2143898](https://doi.org/10.1080/00031305.2022.2143898).

Talks

- Aug. 2026 *Small area estimation of wealth quantiles over time, a Bayesian unit-level modeling approach* JSM 2026, Boston, Massachusetts, USA

[†]Equal contribution.

- Jun. 2026 *Small area estimation of wealth quantiles over time, a Bayesian unit-level modeling approach* SAE 2026, Bucharest, Romania
- Apr. 2026 *Small area estimation of wealth quantiles over time, a Bayesian unit-level modeling approach* SIPP modernization working group, virtual
- Jan. 2026 *Bayesian Unit-level Models for Longitudinal Survey Data under Informative Sampling* U.S. Census Bureau Center for Statistical Research and Methodology Virtual Seminar
- Aug. 2024 *Bayesian Unit-level Modeling of Categorical Survey Data with a Longitudinal Design* Joint Statistical Meetings, (Topic Contributed Session), Portland, Oregon, USA
- Aug. 2023 *Bayesian Unit-level Models for Longitudinal Survey Data under Informative Sampling with Application to the Household Pulse Survey* Joint Statistical Meetings, (Topic Contributed Session), Toronto, Canada

Poster Presentations

- Oct. 2023 *An Approach to Analyzing the Household Pulse Survey using Bayesian Spatio-Temporal Unit-Level Models Under Informative Sampling* with Alex Sun, Paul A. Parker and Scott H. Holan, (Mizzou Statistics 60th Anniversary Conference) University of Missouri
- Sept. 2022 *An Approach to Analyzing the Household Pulse Survey using Bayesian Spatio-Temporal Unit-Level Models Under Informative Sampling* with Alex Sun, Paul A. Parker and Scott H. Holan, (O'Bayes 2022) University of California Santa Cruz

Awards

- 2025 Honorable Mention - ASA SRMS/GSS/SSS Student Paper Competition
- Aug. 2023 - Feb. 2025 U.S. Census Bureau Dissertation Fellowship.
- June 2021 6th Int'l Conference on Establishment Statistics Student Contest winner.

Teaching

- Fall 2020 & Spring 2021 Stat 2500: Introduction to Probability and Statistics I

Service

- 2024-2025 UCSC/Mizzou Modern Survey Statistics Reading Group Coordinator
- 2022-2024 Mizzou Statistics Graduate Student Association Committee Member

Refereed Articles for

The American Statistician
Applied Stochastic Models in Business and Industry
Biometrical Journal
Canadian Journal of Statistics
Computational Statistics & Data Analysis
Journal of Data Science

Statistica

Technical Skills

Languages R, Python, bash

Software Stan, Tidyverse, T_EX, Emacs, vim, SLURM, git, SVN, sed/awk

Languages

English Native

Russian Fluent